Complete monotonicity beyond logarithmic infinite divisibility

A counterexample theorem for Le Roy functions

Keio Research Notes*

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Abstract

For the Le Roy function $F^{(\gamma)}_{\alpha,\beta}(z) = \sum_{n\geq 0} z^n/\Gamma(\beta+\alpha n)^\gamma$, we prove a strict separation between complete monotonicity on the negative half-line and logarithmic infinite divisibility of its representing law. For every fixed real $\gamma > 1$ and $0 < \alpha < 1/\gamma$, let β_0 be the threshold at which $z + \gamma(z^{\beta-\alpha} - z^\beta) \leq 1$ holds for all $0 < z < 1$. There is a nonempty interval immediately below β_0 on which the Le Roy function remains completely monotone. This disproves the universal equivalence proposed by Thomas Simon. The proof combines strict positivity of an infinitely divisible density, obtained from an analytic convolution root, with uniform estimates at both endpoints for an inverse Mellin transform. These estimates allow positivity to persist as the signed Lévy kernel becomes negative. The argument covers nonintegral γ, is independent of numerical computation, and concerns one previously posed conjecture. It does not determine the exact complete monotonicity region.

Keywords. Complete monotonicity; Le Roy function; Mellin inversion; infinite divisibility; positive densities; gamma moments.

1 The original conjecture and the result

A smooth real function h on $(0,\infty)$ is completely monotone (CM) if $(-1)^n h^{(n)}(x) \geq 0$ for all integers $n \geq 0$. A positive random variable is logarithmically infinitely divisible (log-ID) if its logarithm has an infinitely divisible law on $\mathbb{R}$. Consider the entire function

$$F^{(\gamma)}_{\alpha,\beta}(w) = \sum_{n=0}^{\infty} \frac{w^n}{\Gamma(\beta+\alpha n)^\gamma}, \qquad \alpha,\beta,\gamma > 0. \tag{1.1}$$

Simon [1, Section 3] proposed the equivalence

$$F^{(\gamma)}_{\alpha,\beta}(-x) \text{ is CM} \quad \Longleftrightarrow \quad z + \gamma(z^{\beta-\alpha} - z^\beta) \leq 1 \quad \text{for every } 0 < z < 1. \tag{1.2}$$

The inequality is sufficient by a Lévy–Khintchine construction. Its proposed necessity is the target here. The original parameters, function, domain, and universal quantifier in (1.2) are retained.

For fixed $\gamma > 1$, $0 < \alpha < 1/\gamma$, define

$$B(t) = \alpha + \frac{\alpha}{t}\log\left(\frac{\gamma(1-e^{-t})}{1-e^{-t/\alpha}}\right), \qquad \beta_0 = \max_{t>0} B(t). \tag{1.3}$$

The existence of this maximum and its relationship to (1.2) are proved below.

*AI-assisted research manuscript prepared with OpenAI Codex. The argument received separate internal mathematical checks; these are not external peer review. Historical priority is not certified.

Theorem 1.1 (Strict separation)**.** *Fix any real $\gamma > 1$ and $0 < \alpha < 1/\gamma$. Then*

$$\alpha < \beta_0 < \frac{\alpha(1+\gamma)}{2}.$$

There exists $\varepsilon > 0$, with $\beta_0 - \varepsilon > \alpha$, such that for every $\beta \in (\beta_0 - \varepsilon, \beta_0)$:

1. *$F^{(\gamma)}_{\alpha,\beta}(-x)$ is completely monotone on $(0,\infty)$;*
2. *the inequality on the right of (1.2) fails for some $z \in (0,1)$;*
3. *there is a positive random variable X_β, not log-ID, with*

$$\mathbb{E}X_\beta^s = \frac{\Gamma(1+s)\Gamma(\beta)^\gamma}{\Gamma(\beta+\alpha s)^\gamma}, \qquad \operatorname{Re} s > -1, \tag{1.4}$$

and $\mathbb{E}e^{wX_\beta} = \Gamma(\beta)^\gamma F^{(\gamma)}_{\alpha,\beta}(w)$ for every $w \in \mathbb{C}$.

The theorem gives a negative resolution of the literal equivalence (1.2). The interval width is established to be positive; no explicit numerical width or particular rational value of β is asserted. Infinitely many parameter choices remain one conjecture resolution, not multiple independent targets.

The transform representation is classical in the rational/integer specializations studied by Górska, Horzela, and Garrappa [2]. The argument here requires uniformity in β, strict positivity at the log-ID threshold, and continuation across that threshold. These are proved directly, including for noninteger γ.

2 The threshold and the signed Lévy kernel

Fix α, γ as in Theorem 1.1, and write $\nu = 1 - \alpha\gamma \in (0,1)$. For $\beta > \alpha$, put

$$\phi_\beta(t) = \frac{e^{-t}}{1-e^{-t}} - \gamma\frac{e^{-\beta t/\alpha}}{1-e^{-t/\alpha}}, \qquad t > 0. \tag{2.1}$$

Elementary rearrangement gives

$$\phi_\beta(t) \geq 0 \quad \Longleftrightarrow \quad \beta \geq B(t). \tag{2.2}$$

With $z = e^{-t/\alpha}$, the same inequality becomes precisely the right side of (1.2).

Lemma 2.1. *The maximum in (1.3) is attained in $(0,\infty)$, and $\alpha < \beta_0 < \alpha(1+\gamma)/2$. The scalar inequality in (1.2) holds for all $z \in (0,1)$ exactly when $\beta \geq \beta_0$.*

Proof. As $t \downarrow 0$, the logarithm in (1.3) tends to $\log(\alpha\gamma) < 0$, so $B(t) \to -\infty$. As $t \to \infty$,

$$B(t) = \alpha + \frac{\alpha \log\gamma}{t} + o(t^{-1}).$$

Thus B is greater than α somewhere and tends to α at infinity. Continuity implies that its maximum is attained at a finite positive point and exceeds α.

Let $\beta_U = \alpha(1+\gamma)/2$. Since $1/\alpha > \gamma$,

$$\begin{aligned}\phi_{\beta_U}(t) &> \frac{e^{-t}}{1-e^{-t}} - \gamma\frac{e^{-(1+\gamma)t/2}}{1-e^{-\gamma t}} \\ &= \frac{e^{-t/2}}{2}\left\{\frac{1}{\sinh(t/2)} - \frac{\gamma}{\sinh(\gamma t/2)}\right\} > 0.\end{aligned}$$

The last inequality follows because $u \mapsto \sinh(u)/u$ is strictly increasing for $u > 0$. Hence $B(t) < \beta_U$ at every positive t. Since its maximum is attained, $\beta_0 < \beta_U$. The equivalence follows from (2.2). □

For complex s with $\operatorname{Re} s > -1$, define

$$M_\beta(s) = \Gamma(1+s)\Gamma(\beta)^\gamma \exp\{-\gamma \operatorname{Log}\Gamma(\beta+\alpha s)\}. \tag{2.3}$$

Here $\operatorname{Log}\Gamma$ is the holomorphic logarithm on the right half-plane that is real on the positive real axis. This convention is essential when $\gamma \notin \mathbb{N}$. It is available because $\operatorname{Re}(\beta+\alpha s) > \beta - \alpha > 0$.

The logarithmic gamma integral gives

$$\log M_\beta(s) = d_\beta s + \int_0^\infty (e^{-st} - 1 + st)\phi_\beta(t)\,\frac{\mathrm{d}t}{t}, \qquad d_\beta = \psi(1) - \alpha\gamma\psi(\beta). \tag{2.4}$$

For clarity, this identity can be checked without choosing an implicit complex power. Twice differentiating both sides gives $\psi'(1+s) - \alpha^2\gamma\psi'(\beta+\alpha s)$, using the standard integral for the trigamma function. The two sides agree at $s = 0$ and have the same derivative there. Analyticity proves the identity throughout $\operatorname{Re} s > -1$.

At the endpoints,

$$\phi_\beta(t) = \frac{\nu}{t} + O(1) \quad (t \downarrow 0), \qquad \phi_\beta(t) = O(e^{-t}) \quad (t \to \infty). \tag{2.5}$$

Consequently $\phi_\beta(t)\,\mathrm{d}t/t$ is a signed Lévy measure with finite absolute second moment. At $\beta = \beta_0$ it is nonnegative, so (2.4) with $s = \mathrm{i}u$ is the characteristic exponent of an infinitely divisible law. Its jumps correspond to $-t$, and the compensation $+\mathrm{i}ut$ in (2.4) is admissible by (2.5).

3 An auxiliary positivity-continuation problem

The useful auxiliary question is the following: when can a family of inverse transforms remain nonnegative after its logarithmic transform loses a nonnegative Lévy representation? Two elementary facts give a criterion. They are stated explicitly to separate the general argument from its special-function estimates.

Lemma 3.1 (Strict positivity from a convolution root)**.** *Let χ be an infinitely divisible characteristic function on $\mathbb{R}$. Suppose*

$$|\chi(u)| \le C(1+|u|)^A e^{-a|u|} \quad (u \in \mathbb{R})$$

for some $a, C > 0$ and finite A. Its probability density is continuous and strictly positive at every real point.

Proof. Infinite divisibility supplies a characteristic function θ with $\theta^2 = \chi$, and

$$|\theta(u)| \le C^{1/2}(1+|u|)^{A/2} e^{-a|u|/2}.$$

Fourier inversion gives a bounded continuous nonnegative probability density p for this convolution root. The same integral defines a holomorphic function in $|\operatorname{Im} w| < a/2$. Thus p is real analytic and not identically zero. Its real zeros are isolated, so $p(v)p(w-v) > 0$ outside a discrete set for every fixed real w. It follows that

$$(p * p)(w) = \int_{\mathbb{R}} p(v)p(w-v)\,\mathrm{d}v > 0.$$

This integral is finite since p is bounded and integrable. The convolution is the continuous density with characteristic function χ. □

Lemma 3.2 (Continuation with uniform endpoint control)**.** *Let $q_\theta(x)$ be jointly continuous for $x > 0$ and real θ in a neighborhood of θ_0. Suppose $q_{\theta_0}(x) > 0$ for every $x > 0$. If there are $0 < x_0 < x_1 < \infty$ such that*

$$q_\theta(x) > 0 \quad \text{on } (0, x_0] \cup [x_1, \infty)$$

uniformly for θ in that neighborhood, then $q_\theta(x) > 0$ for all $x > 0$ whenever $|\theta - \theta_0|$ is sufficiently small.

Proof. The continuous function q_{θ_0} has a positive minimum on $[x_0, x_1]$. Uniform continuity on a compact parameter neighborhood times this interval preserves, for nearby θ, at least half that minimum. The assumed endpoint control covers the rest of the positive axis. □

This compactness argument alone does not prove the theorem: pointwise positivity or pointwise endpoint asymptotics would not supply its uniform hypothesis. The next section establishes precisely that hypothesis for (2.3).

4 Inverse Mellin densities and uniform tails

Let K be any compact interval in (α, ∞). All constants and uniform remainders in this section may depend on α, γ, K. Define the real-valued candidate density

$$q_\beta(x) = \frac{1}{2\pi \mathrm{i} x} \int_{-\mathrm{i}\infty}^{\mathrm{i}\infty} M_\beta(s) x^{-s} \, \mathrm{d}s, \qquad x > 0, \quad \beta \in K. \tag{4.1}$$

It need not initially be nonnegative. Stirling's formula on each fixed vertical line supplies a polynomial bound times $e^{-\pi\nu|\operatorname{Im} s|/2}$, uniformly for $\beta \in K$. The integral therefore converges absolutely and is jointly continuous in (β, x). Reality follows by complex conjugation.

4.1 The endpoint at zero

Choose

$$0 < \eta < \min\{1, (\min K)/\alpha - 1\}. \tag{4.2}$$

On the whole strip $\operatorname{Re} s \geq -1-\eta$, the denominator in (2.3) uses a single holomorphic logarithm, because $\operatorname{Re}(\beta + \alpha s) > 0$. Shifting the contour in (4.1) to $\operatorname{Re} s = -1 - \eta$ crosses only the simple numerator pole at $s = -1$. The horizontal integrals tend to zero by the vertical Stirling bounds. Taking the residue yields

$$q_\beta(x) = A_\beta + O_K(x^\eta), \qquad A_\beta = \frac{\Gamma(\beta)^\gamma}{\Gamma(\beta - \alpha)^\gamma} > 0, \qquad x \downarrow 0. \tag{4.3}$$

The remainder is uniform because the shifted integral is bounded absolutely and uniformly after factoring out x^η. In particular, $\min_{\beta \in K} A_\beta > 0$.

4.2 The endpoint at infinity

Introduce

$$\kappa_\beta = \frac{1+\gamma}{2} - \gamma\beta, \qquad C_\beta = (2\pi)^{(1-\gamma)/2} \Gamma(\beta)^\gamma \alpha^{\gamma(1/2-\beta)}, \tag{4.4}$$

$$r = (\alpha^{\alpha\gamma} x)^{1/\nu}. \tag{4.5}$$

Uniform sectorial Stirling estimates [6, Section 5.11] give, as $|s| \to \infty$ in the right half-plane,

$$M_\beta(s) = C_\beta s^{\kappa_\beta} \exp\{\nu s(\log s - 1) - \alpha\gamma s \log \alpha\}(1 + O_K(|s|^{-1})). \tag{4.6}$$

Here $\log s$ is the principal logarithm in that half-plane. The shifted gamma arguments vary in a compact family, so the remainder is uniform in β. Taking the fixed real power γ through the holomorphic logarithm preserves this estimate.

Proposition 4.1 (Uniform saddle estimate)**.** *Uniformly for $\beta \in K$,*

$$q_\beta(x) = \frac{C_\beta}{x\sqrt{2\pi\nu}} r^{\kappa_\beta+1/2} e^{-\nu r}(1 + o_K(1)), \qquad x \to \infty. \tag{4.7}$$

Proof. Move the contour in (4.1) to $\operatorname{Re} s = r$. No poles are crossed; for each fixed r, horizontal integrals vanish as their imaginary height tends to infinity. Because $\log x + \alpha\gamma \log \alpha = \nu \log r$, the exponential part of $M_\beta(s)x^{-s}$ is

$$\exp\{\nu s(\log(s/r) - 1)\}.$$

Set $s = r + \mathrm{i}\sqrt{r}\, u$. Expansion around $s = r$ gives

$$\frac{M_\beta(r + \mathrm{i}\sqrt{r}\, u)x^{-r-\mathrm{i}\sqrt{r}\, u}}{C_\beta r^{\kappa_\beta} e^{-\nu r}} \longrightarrow e^{-\nu u^2/2} \tag{4.8}$$

for each fixed u, uniformly for $\beta \in K$. In detail, the first derivative of the exponent vanishes, its second derivative is ν/r, and multiplication by $(\mathrm{i}\sqrt{r}\, u)^2/2$ produces $-\nu u^2/2$.

To control the entire contour, let $v = u/\sqrt{r}$ and define

$$H(v) = \tfrac{1}{2}\log(1 + v^2) - v \arctan v.$$

The modulus of the left side of (4.8) is bounded by

$$D_K(1 + v^2)^{\kappa_\beta/2} e^{\nu r H(v)}. \tag{4.9}$$

The function H is even, $H(0) = 0$, and $H'(v) = -\arctan v$ for $v > 0$. Hence

$$H(v) \le -v^2/4 \qquad (|v| \le 1), \tag{4.10}$$

$$H(v) \le H(1) - \tfrac{\pi}{4}(|v| - 1) \qquad (|v| \ge 1), \tag{4.11}$$

with $H(1) < 0$. The factor involving κ_β is uniformly bounded on $|v| \le 1$; this part of the integrand is dominated by a constant times $e^{-\nu u^2/4}$.

For the outer region put $\kappa_+ = \max(0, \max_{\beta\in K} \kappa_\beta)$. Since $\log(1 + v^2) \le \log 2 + (|v| - 1)$ for $|v| \ge 1$, (4.9)–(4.11), for all sufficiently large r, are bounded by

$$D'_K e^{\nu r H(1)} \exp\{-\nu\pi r(|v| - 1)/8\}.$$

Its integral over $|u| \ge \sqrt{r}$ is $O_K(r^{-1/2} e^{\nu r H(1)})$. Thus the outer contour is uniformly negligible. Dominated convergence on the central region, uniform on bounded u-intervals in β, now gives the uniform integral limit $\int_{\mathbb{R}} e^{-\nu u^2/2}\, \mathrm{d}u = \sqrt{2\pi/\nu}$. Finally $\mathrm{d}s = \mathrm{i}\sqrt{r}\, \mathrm{d}u$ and the prefactor $1/(2\pi \mathrm{i} x)$ in (4.1) yield (4.7). □

All leading factors in (4.3) and (4.7) are positive. Therefore, for this entire compact interval K, there are $0 < x_0 < x_1 < \infty$ such that

$$q_\beta(x) > 0 \quad (\beta \in K,\ x \in (0, x_0] \cup [x_1, \infty)). \tag{4.12}$$

This is a uniform conclusion on an unbounded domain, not an inference from a finite numerical grid.

4.3 Normalization before positivity

The endpoint estimates imply absolute integrability of $x^s q_\beta(x)$ for every $\operatorname{Re} s > -1$. Let $p_\beta(y) = e^y q_\beta(e^y)$. The definition (4.1) says that p_β is the inverse Fourier transform of the integrable function $u \mapsto M_\beta(\mathrm{i}u)$. The endpoint bounds also give $p_\beta \in L^1(\mathbb{R})$. Fourier inversion therefore identifies its Fourier transform with $M_\beta(\mathrm{i}u)$. Both sides of

$$\int_0^\infty x^s q_\beta(x)\,\mathrm{d}x = M_\beta(s) \tag{4.13}$$

are holomorphic on $\operatorname{Re} s > -1$, so the identity theorem extends this equality from the imaginary axis to that half-plane. In particular,

$$\int_0^\infty q_\beta(x)\,\mathrm{d}x = 1. \tag{4.14}$$

No nonnegativity assumption was used in this normalization.

5 Proof of strict separation

Choose a compact interval $K \subset (\alpha, \infty)$ with β_0 in its interior. By Section 2, $M_{\beta_0}(\mathrm{i}u)$ is an infinitely divisible characteristic function. Vertical Stirling estimates give

$$|M_{\beta_0}(\mathrm{i}u)| \le D(1+|u|)^A e^{-\pi\nu|u|/2}$$

for some finite A and $D > 0$. Lemma 3.1 shows that its density is strictly positive on $\mathbb{R}$. Fourier uniqueness identifies that density with $e^y q_{\beta_0}(e^y)$, so

$$q_{\beta_0}(x) > 0 \qquad (x > 0). \tag{5.1}$$

This step uses infinite divisibility only at the threshold itself.

Joint continuity, (4.12), and Lemma 3.2 imply that q_β remains strictly positive everywhere for $|\beta - \beta_0| < \varepsilon$, with some $\varepsilon > 0$ small enough that this interval lies in K. By (4.14), these functions are probability densities. Let X_β have density q_β. Equation (4.13) proves (1.4).

Because $0 < \nu < 1$, the stretched exponential in (4.7) decays faster than e^{-cx} for every fixed $c > 0$. Thus $\mathbb{E}e^{cX_\beta} < \infty$ for all $c > 0$, and the exponential series can be integrated absolutely for every $w \in \mathbb{C}$. It follows that

$$\mathbb{E}e^{wX_\beta} = \sum_{n\ge 0} \frac{w^n}{n!} M_\beta(n) = \Gamma(\beta)^\gamma F^{(\gamma)}_{\alpha,\beta}(w), \tag{5.2}$$

$$F^{(\gamma)}_{\alpha,\beta}(-x) = \Gamma(\beta)^{-\gamma}\mathbb{E}e^{-xX_\beta}. \tag{5.3}$$

Differentiating the Laplace integral proves complete monotonicity. For $\beta < \beta_0$, Lemma 2.1 proves failure of the unchanged scalar inequality in (1.2).

It remains to justify explicitly the stronger non-log-ID conclusion. For such a β, continuity and (2.2) show that ϕ_β is negative on a nonempty open interval. Let $V = \log X_\beta$. Its characteristic function is $M_\beta(\mathrm{i}u)$, and (2.4) is its continuous characteristic logarithm, equal to zero at $u = 0$. Twice differentiating yields

$$-\frac{\mathrm{d}^2}{\mathrm{d}u^2}\log M_\beta(\mathrm{i}u) = \int_0^\infty e^{-\mathrm{i}ut}\, t\phi_\beta(t)\,\mathrm{d}t. \tag{5.4}$$

The signed measure on the right has finite total variation by (2.5). Also $\mathbb{E}V^2 < \infty$, directly from (4.3) and (4.7).

If V were infinitely divisible, its finite variance would give a Gaussian coefficient $\sigma^2 \geq 0$ and a nonnegative Lévy measure Π with $\int y^2 \Pi(\mathrm{d}y) < \infty$. The left side of (5.4) would then be the Fourier transform of the finite nonnegative measure

$$\sigma^2 \delta_0(\mathrm{d}y) + y^2 \Pi(\mathrm{d}y).$$

Uniqueness of Fourier transforms of finite measures identifies this measure with the pushforward of $t\phi_\beta(t)\,\mathrm{d}t$ under $t \mapsto -t$. The latter is negative on an interval, a contradiction. This completes the proof of Theorem 1.1. □

6 A concrete specialization and consistency checks

For $\alpha = 1/3$ and $\gamma = 2$, setting $z = t^3$ in the scalar inequality and dividing by $1 - t$ reduces it to

$$2t^{3\beta-1} \leq 1 + t + t^2 \qquad (0 < t < 1).$$

Thus

$$b_* := \beta_0(1/3, 2) = \frac{1}{3}\left(1 + \max_{0<t<1} \frac{\log((1 + t + t^2)/2)}{\log t}\right). \tag{6.1}$$

The quotient tends to zero from above as $t \downarrow 0$ and to $-\infty$ as $t \uparrow 1$. Its maximum is attained. Furthermore

$$1 + t + t^2 - 2\sqrt{t} = (1 - \sqrt{t})^2 + t^2 > 0$$

implies $1/3 < b_* < 1/2$. Numerical evaluation gives $b_* \approx 0.4347490386$; this decimal is illustrative and is not used to certify any particular parameter as a counterexample.

For an independent check of the constants, let $Z = X^3/27$. The gamma triplication formula gives the inverse Mellin density

$$f_\beta(z) = \frac{\Gamma(\beta)^2}{\Gamma(1/3)\Gamma(2/3)} G^{3,0}_{2,3}\left(z \,\middle|\, \begin{matrix} \beta - 1, \beta - 1 \\ -2/3, -1/3, 0 \end{matrix}\right). \tag{6.2}$$

Here the Meijer function may simply be defined by the absolutely convergent contour integral

$$\frac{1}{2\pi \mathrm{i}} \int_{1-\mathrm{i}\infty}^{1+\mathrm{i}\infty} \frac{\Gamma(s - 2/3)\Gamma(s - 1/3)\Gamma(s)}{\Gamma(s + \beta - 1)^2} z^{-s}\,\mathrm{d}s.$$

The transformation of densities is $f_\beta(z) = z^{-2/3} q_\beta(3z^{1/3})$. Accordingly, (4.3) and (4.7) give

$$f_\beta(z) = \frac{\Gamma(\beta)^2}{\Gamma(\beta - 1/3)^2} z^{-2/3}(1 + o_K(1)), \qquad z \downarrow 0, \tag{6.3}$$

$$f_\beta(z) = \frac{\Gamma(\beta)^2}{\Gamma(1/3)\Gamma(2/3)} z^{1-2\beta} e^{-z}(1 + o_K(1)), \qquad z \to \infty. \tag{6.4}$$

These agree with the special contour calculation. The general proof, however, needs neither rational α nor integer γ.

As a further normalization check on the saddle constants, the formulas with $\alpha = 1/2, \gamma = 1, \beta = 1$ give $q(x) = \pi^{-1/2} e^{-x^2/4}$, the known half-normal density. This check is outside the counterexample regime and is not an additional conjecture resolution.

7 Prior work, the corpus, and the scope of the claim

Classical inputs and the proposed contribution. Mellin and Fourier inversion, gamma asymptotics, Lévy–Khintchine theory, and compactness are classical. Górska, Horzela, and Garrappa [2] obtained inverse Meijer representations and pointwise asymptotics in rational/integer cases. Their numerical positivity thresholds do not provide the uniform continuation proof above. The contribution proposed here is the strict gap theorem, established by the analytic convolution-root argument together with uniform endpoint control for all real parameters in the stated regime.

Gamma-type probability distributions with non-ID logarithms are not new in general. Dufresne [3, Example 2] gives such an example, and Kadankova, Simon, and Wang [4] construct related signed spectral measures. Wang [5] studies further gamma moment and quasi-infinite-divisibility questions. The theorem here concerns the specific family and universal equivalence (1.2); it does not claim the first gamma-type non-ID law.

How the supplied corpus was used. The Verifiable Mathematical Reasoning Corpus [7] was examined at commit `ecd87de3348dcb706a76bb12fde722d7a348a012`. Manuscript [067], especially its method M9, suggests passing from derivative signs to Laplace-density signs. Manuscript [096] emphasizes exact integral representations, endpoint analysis, and separation of classical inputs from a new application. Those ideas motivated the auxiliary problem in Section 3. No unverified theorem from the corpus is a premise of the proof; the required estimates and positivity arguments are given in full here.

Priority and verification. A targeted literature search through 3 November 2026 examined the original paper, its relevant gamma-moment references, the earlier Meijer representations, later gamma-moment work, and exact-title and conjecture searches. It did not locate an identical resolution of (1.2). This is a report about the search performed, not a proof of the absence of prior work. Historical priority and external peer validation remain unconfirmed. The separate question concerning convexity of Simon's scalar threshold is not claimed as a result here.

The proof was checked in separate internal derivations of the $(1/3, 2)$ case and the full real-parameter case. The checks covered branch choices, residues, uniform saddle bounds, normalization, strict positivity, and signed Lévy uniqueness. Numerical samples were exploratory only. No finite search, assumed positivity theorem, or unproved conjecture is needed in the proof.

Research count and remaining work. This manuscript contains a complete analytic argument against one original conjecture. The broader request for ten difficult conjecture resolutions has not been achieved. Other investigated problems in number theory, graph theory, and symbolic integration produced no additional unrestricted resolution. Restricted classes, already published solutions, and multiple instances of the same family have not been counted. The exact CM threshold for Le Roy functions remains undetermined by this paper, as does a certified numerical width of the counterexample interval.

References

[1] T. Simon, *Remark on a Mittag–Leffler function of Le Roy type*, Integral Transforms and Special Functions **33** (2022), no. 2, 108–114. doi:10.1080/10652469.2021.1913138. Author version: arXiv:2012.11018v2, 25 March 2021; conjecture in Section 3.

[2] K. Górska, A. Horzela, and R. Garrappa, *Some results on the complete monotonicity of the Mittag–Leffler functions of Le Roy type*, Fractional Calculus and Applied Analysis **22** (2019), no. 5, 1284–1306. doi:10.1515/fca-2019-0068. arXiv:1912.07100v2.

[3] D. Dufresne, *G distributions and the beta-gamma algebra*, Electronic Journal of Probability **15** (2010), no. 71, 2163–2199. doi:10.1214/EJP.v15-845.

[4] T. Kadankova, T. Simon, and M. Wang, *On some new moments of Gamma type*, Statistics & Probability Letters **165** (2020), 108854. doi:10.1016/j.spl.2020.108854. arXiv:1908.03428.

[5] M. Wang, *Moments of Gamma type and three-parametric Mittag–Leffler function*, preprint, 2024. arXiv:2410.19330.

[6] NIST Digital Library of Mathematical Functions, *Section 5.11: Asymptotic Expansions*, `https://dlmf.nist.gov/5.11`. Accessed 3 November 2026.

[7] QEEDDD, *Verifiable Mathematical Reasoning Corpus*, GitHub repository, snapshot of 2 November 2026, commit `ecd87de3348dcb706a76bb12fde722d7a348a012`. `https://github.com/QEEDDD/Verifiable-Mathematical-Reasoning-Corpus`. Methodological comparisons: manuscripts [067] and [096].